

91. Alternation (OR): $|$

Alternation regexp me simple "OR" hai, vertical line $|$ se.

Programming languages dhundhne hain (HTML, PHP, Java ya JavaScript):

```
html|php|java(script)?
```

```
js
```

```
let regexp = /html|php|css|java(script)?/gi;  
"First HTML appeared, then CSS, then JavaScript".match(regexp); // 'HTML', 'C'
```

Square brackets bhi choose karne dete hain ($gr[ae]y$) = $gray$ ya $grey$), par wo sirf **characters ya character classes** allow karte hain. Alternation **koi bhi expressions** allow karta hai. $A|B|C$ = A , B ya C me se koi ek expression.

- $gr(a|e)y$ = $gr[ae]y$
- $gra|ey$ = gra ya ey

Pattern ke chune hue hisse par alternation lagane ke liye **parentheses**:

- $I\ love\ HTML|CSS$ = $I\ love\ HTML$ ya CSS
- $I\ love\ (HTML|CSS)$ = $I\ love\ HTML$ ya $I\ love\ CSS$

Example: time ke liye regexp

Simple $\backslash d\backslash d:\backslash d\backslash d$ bahut dheela hai ($25:99$) bhi accept karta hai). Behtar:

- Hours:** pehla digit 0 ya 1 ho to agla koi bhi digit ($[01]\backslash d$); pehla digit 2 ho to agla $[0-3]$. Alternation: $[01]\backslash d|2[0-3]$.
- Minutes:** 00 se 59 : $[0-5]\backslash d$.

Seedha jodne par $[01]\backslash d|2[0-3]:[0-5]\backslash d$ galat hai: alternation $[01]\backslash d$ ya $2[0-3]:[0-5]\backslash d$ ke beech ho jaata hai (minutes sirf doosre variant me judte hain). Hours ko parentheses me lo:

```
js
```

```
let regexp = /([01]\d|2[0-3]):[0-5]\d/g;  
"00:00 10:10 23:59 25:99 1:2".match(regexp); // 00:00,10:10,23:59
```

Tasks

Programming languages ($Java\ JavaScript\ PHP\ C++\ C$): seedha

$Java|JavaScript|PHP|C|C++|C$ **galat** hai (result $Java,Java,PHP,C,C$), kyunki engine

alternatives ek ke baad ek check karta hai: pehle `Java` mil jaata hai to `JavaScript` kabhi nahi milta, `C` pehle to `C++` nahi. Do fix:

1. **Lambe variants pehle:** `JavaScript|Java|C\+\+|C|PHP`
2. **Same shuruaat wale variants merge:** `Java(Script)?|C(\+\+)?|PHP`

BB-tag pairs (`[b]...[/b]`, `[url]...[/url]`, `[quote]...[/quote]`), **nested ho sakte hain, newlines bhi:** `/\[ (b|url|quote) ] .*? \[ \[ \1 \] \] /gs`. Opening tag `\[ (b|url|quote) ]`, lazy `. * ?` (`s` flag se newline bhi), aur closing tag backreference `\1` se. Closing tag ka slash bhi escape karna padta hai.

Quoted strings `"..."` (escaped quotes `\"`, `\\`, `\n` ke saath): `/"(\\.|[^\n\\])*"/g`

- Opening quote `"`
- Backslash ho to uske baad koi bhi character (`\\.`)
- Warna quote aur backslash ke alawa koi bhi character (`[^\n\\]`)
- ... closing quote tak

Poora `<style...>` tag (par `<styler>` nahi): `<style(>|\s.*?>)`. Yaani `<style` ke baad ya to seedha `>` ya space aur phir kuch bhi `>` tak.